Template T = (O, X, C, Pre, P, π, Σ, Ĵ, Γ)

one new object O, its decisions X (program + chip),
the next-level program P, and a cost model Ĵ

one object, read three ways

Search face → solver

domain X and constraints C
define the feasible region

min Ĵ(x) s.t. x ∈ X, C(x)

MILP / SMT; returns an assignment,
a bound, or infeasibility

Scope face → agent, population

Γ: workloads and chip contracts
where the template may be reused;

Σ: signature for diversity and retrieval

Correctness face → verifier

preconditions Pre(x)
backward map π: level ℓ+1 → ℓ
certificate (R, ρ) for the
four refinement obligations

proved once for the whole family;
each instance only checks Pre(x)

Contract conditions

K1 C(x) ⇒ Pre(x): every feasible point is admissible

K2 Pre(x) ⇒ the refinement is sound (proved once)

K3 C is satisfiable within Γ

Instance: GEMM → shared-memory tiling with an S-stage pipeline

Search constraints C

$S \cdot e \cdot (Tm \cdot Tk + Tk \cdot Tn) \le M_s$ (shared-memory capacity)

$Tm \cdot Tn \le R_{acc}$ (accumulator registers)

$P \le S - 2 + \beta$ (buffer lifetime)

tiles Tm,Tn,Tk · stages S · prefetch distance P · trailing barrier β ∈ {0,1}

K1 →

Obligations discharged by the verifier

index: coverage · exact-once · no phantom accesses

schedule: readiness (no read before load completes)

lifetime (no slot overwritten while still being read)

$P \le S - 2 + \beta$ is exact: it matches the checker on 24/24 schedule cases